

# **LAPORAN PRAKTIKUM**

## **TEKNOLOGI IOT 2**

Fondasi Arsitektur Internet of Things (IoT) dan Mikrokontroler



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## Packet Tracer - Who Hears the Broadcast?

### Objectives

**Part 1: Observe Broadcast Traffic in a VLAN Implementation Part**

**2: Complete Review Questions**

### Scenario

In this activity, a 24-port Catalyst 2960 switch is fully populated. All ports are in use. You will observe broadcast traffic in a VLAN implementation and answer some reflection questions.

### Instructions

#### Step 1: Use ping to generate traffic.

- a. Click **PC0** and click the **Desktop** tab> **Command Prompt**.
- b. Enter the **ping 192.168.1.8** command. The ping should succeed.

Unlike a LAN, a VLAN is a broadcast domain created by switches. Using Packet Tracer **Simulation** mode, ping the end devices within their own VLAN. Based on your observation, answer the questions in Step 2.

#### Step 2: Generate and examine broadcast traffic in a VLAN implementation. a.

Switch to **Simulation** mode.

- b. Click **Edit Filters** in the Simulation Panel. Uncheck the **Show All/None** checkbox. Check the **ICMP** checkbox.
- c. Click the **Add Complex PDU** tool, this is the open envelope icon on the right toolbar.
- d. Float the mouse cursor over the topology and the pointer changes to an envelope with a plus (+) sign.
- e. Click **PC0** to serve as the source for this test message and the **Create Complex PDU** dialog window opens. Enter the following values:

o Destination IP Address: 255.255.255.255 (broadcast address) o Sequence

Number: 1 o One Shot Time: 0

Within the PDU settings, the default for **Select Application:** is PING.

What are at least 3 other applications available for use?

Tiga aplikasi lain yang tersedia antara lain:

1. HTTP
  2. DNS
  3. FTP
- f. Click **Create PDU**. This test broadcast packet now appears in the **Simulation Panel Event List**. It also appears in the PDU List window. It is the first PDU for Scenario 0.

- g. Click **Capture/Forward** twice.

What happened to the packet?

Paket broadcast dikirim ke semua perangkat yang berada dalam **VLAN yang sama**. Perangkat yang berada di VLAN berbeda tidak menerima paket tersebut. Switch meneruskan broadcast hanya dalam broadcast domain/VLAN yang sama.

- h. Repeat this process for **PC8** and **PC16**.

## Packet Tracer - Who Hears the Broadcast?

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### Reflection Questions

1. If a PC in VLAN 10 sends a broadcast message, which devices receive it?

Broadcast akan diterima oleh semua perangkat yang berada dalam **VLAN 10** pada switch tersebut. Perangkat yang berada di VLAN 20 dan VLAN 30 tidak menerima broadcast tersebut.

2. If a PC in VLAN 20 sends a broadcast message, which devices receive it?

Broadcast akan diterima oleh semua perangkat yang berada dalam **VLAN 20**. Perangkat yang berada di VLAN 10 dan VLAN 30 tidak menerima broadcast tersebut.

3. If a PC in VLAN 30 sends a broadcast message, which devices receive it?

Broadcast akan diterima oleh semua perangkat yang berada dalam **VLAN 30**. Perangkat yang berada di VLAN 10 dan VLAN 20 tidak menerima broadcast tersebut.

4. What happens to a frame sent from a PC in VLAN 10 to a PC in VLAN 30?

Frame tidak dapat berkomunikasi secara langsung karena VLAN 10 dan VLAN 30 merupakan **broadcast domain yang berbeda**. Diperlukan perangkat Layer 3 seperti **router atau multilayer switch** untuk melakukan komunikasi antar-VLAN.

5. Which ports on the switch light up if a PC connected to port 11 sends a unicast message to a PC connected to port 13?

**Port 11 dan port 13** yang terlibat dalam pengiriman frame. Karena merupakan komunikasi **unicast**, switch hanya meneruskan frame ke port tujuan yang sesuai, bukan ke semua port

6. Which ports on the switch light if a PC connected to port 2 sends a unicast message to a PC connected to port 23?

Jika port 2 dan port 23 berada pada **VLAN yang berbeda**, frame tidak dapat langsung diteruskan dari port 2 ke port 23. Komunikasi antar-VLAN membutuhkan perangkat Layer 3.

7. In terms of ports, what are the collision domains on the switch?

Setiap **port switch merupakan satu collision domain**. Jadi, jika switch memiliki 24 port yang aktif, terdapat **24 collision domain**.

8. In terms of ports, what are the broadcast domains on the switch?

Setiap **VLAN merupakan satu broadcast domain**. Jadi, jumlah broadcast domain bergantung pada jumlah VLAN yang dikonfigurasi pada switch.

## Packet Tracer - Investigate a VLAN Implementation

### Addressing Table

| Device | Interface | IP Address   | Subnet Mask   | Default Gateway |
|--------|-----------|--------------|---------------|-----------------|
| S1     | VLAN 99   | 172.17.99.31 | 255.255.255.0 | N/A             |
| S2     | VLAN 99   | 172.17.99.32 | 255.255.255.0 | N/A             |
| S3     | VLAN 99   | 172.17.99.33 | 255.255.255.0 | N/A             |
| PC1    | NIC       | 172.17.10.21 | 255.255.255.0 | 172.17.10.1     |
| PC2    | NIC       | 172.17.20.22 | 255.255.255.0 | 172.17.20.1     |
| PC3    | NIC       | 172.17.30.23 | 255.255.255.0 | 172.17.30.1     |
| PC4    | NIC       | 172.17.10.24 | 255.255.255.0 | 172.17.10.1     |
| PC5    | NIC       | 172.17.20.25 | 255.255.255.0 | 172.17.20.1     |
| PC6    | NIC       | 172.17.30.26 | 255.255.255.0 | 172.17.30.1     |
| PC7    | NIC       | 172.17.10.27 | 255.255.255.0 | 172.17.10.1     |
| PC8    | NIC       | 172.17.20.28 | 255.255.255.0 | 172.17.20.1     |
| PC9    | NIC       | 172.17.30.29 | 255.255.255.0 | 172.17.30.1     |

### Objectives

- Part 1: Observe Broadcast Traffic in a VLAN Implementation**
- Part 2: Observe Broadcast Traffic without VLANs**

### Background

In this activity, you will observe how broadcast traffic is forwarded by the switches when VLANs are configured and when VLANs are not configured.

## Instructions Part 1: Observe Broadcast Traffic in a VLAN Implementation

### Step 1: Ping from PC1 to PC6.

- Wait for all the link lights to turn to green. To accelerate this process, click **Fast Forward Time** located in the bottom tool bar.
- Click the **Simulation** tab and use the **Add Simple PDU** tool. Click **PC1**, and then click **PC6**.
- Click the **Capture/Forward** button to step through the process. Observe the ARP requests as they traverse the network. When the Buffer Full window appears, click the **View Previous Events** button.

Were the pings successful? Explain.

## Packet Tracer - Investigate a VLAN Implementation

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Look at the Simulation Panel, where did **S3** send the packet after receiving it?

In normal operation, when a switch receives a broadcast frame on one of its ports, it forwards the frame out all other ports. Notice that **S2** only sends the ARP request out F0/1 to **S1**. Also notice that **S3** only sends the ARP request out F0/11 to **PC4**. **PC1** and **PC4** both belong to VLAN 10. **PC6** belongs to VLAN 30. Because broadcast traffic is contained within the VLAN, **PC6** never receives the ARP request from **PC1**. Because **PC4** is not the destination, it discards the ARP request. The ping from **PC1** fails because **PC1** never receives an ARP reply.

### Step 2: Ping from PC1 to PC4.

- Click the **New** button under the Scenario 0 dropdown tab. Now click on the **Add Simple PDU** icon on the right side of Packet Tracer and ping from **PC1** to **PC4**.
- Click the **Capture/Forward** button to step through the process. Observe the ARP requests as they traverse the network. When the Buffer Full window appears, click the **View Previous Events** button.

Were the pings successful? Explain.

**No, the ping was unsuccessful.** PC1 dan PC6 berada di VLAN yang berbeda. PC1 berada di VLAN 10, sedangkan PC6 berada di VLAN 30. Broadcast ARP dari PC1 hanya diteruskan di dalam VLAN 10, sehingga PC6 tidak menerima ARP request dan PC1 tidak mendapatkan ARP reply. c. Examine the Simulation Panel.

When the packet reached **S1**, why does it also forward the packet to **PC7**?

Karena PC4 berada di VLAN 10, sedangkan PC6 berada di VLAN 30

Part 2:

## Observe Broadcast Traffic without VLANs

### Step 1: Clear the configurations on all three switches and delete the VLAN database. a.

Return to **Realtime** mode.

- Delete the startup configuration on all 3 switches.

What command is used to delete the startup configuration of the switches?

Where is the VLAN file stored in the switches?

**Yes, the ping was successful.** PC1 dan PC4 sama-sama berada di VLAN 10, sehingga broadcast ARP dari PC1 dapat diteruskan melalui VLAN 10 sampai ke PC4. PC4 kemudian memberikan ARP reply sehingga komunikasi dapat dilakukan.

- c. Delete the VLAN file on all 3 switches.

What command deletes the VLAN file stored in the switches?

Because **PC7 is also a member of VLAN 10**. A switch forwards broadcast frames to all ports that belong to the same VLAN, except the port where the frame was received.

### Step 2: Reload the switches.

Use the **reload** command in privileged EXEC mode to reset all the switches. Wait for the entire link to turn green. To accelerate this process, click **Fast Forward Time** located in the bottom yellow tool bar.

### Step 3: Click Capture/Forward to send ARP requests and pings.

- a. After the switches reload and the link lights return to green, the network is ready to forward your ARP and ping traffic.
- b. Select **Scenario 0** from the drop-down tab to return to Scenario 0.

### Packet Tracer - Investigate a VLAN Implementation

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- c. From **Simulation** mode, click the **Capture/Forward** button to step through the process. Notice that the switches now forward the ARP requests out all ports, except the port on which the ARP request was received. This default action of switches is why VLANs can improve network performance. Broadcast traffic is contained within each VLAN. When the **Buffer Full** window appears, click the **View Previous Events** button.

### Reflection Questions

1. If a PC in VLAN 10 sends a broadcast message, which devices receive it?

Semua perangkat yang berada di **VLAN 10**, yaitu PC1, PC4, dan PC7.

2. If a PC in VLAN 20 sends a broadcast message, which devices receive it?

Semua perangkat yang berada di **VLAN 20**.

3. If a PC in VLAN 30 sends a broadcast message, which devices receive it?

Semua perangkat yang berada di **VLAN 30**.

4. What happens to a frame sent from a PC in VLAN 10 to a PC in VLAN 30?

Frame tidak dapat dikirim langsung karena VLAN 10 dan VLAN 30 merupakan jaringan yang berbeda. Diperlukan **router atau switch Layer 3** untuk komunikasi antar-VLAN.

5. In terms of ports, what are the collision domains on the switch?

Setiap **port switch merupakan satu collision domain**.

6. In terms of ports, what are the broadcast domains on the switch?

Setiap **VLAN merupakan satu broadcast domain**. Pada topologi ini terdapat VLAN 10, VLAN 20, dan VLAN 30, sehingga terdapat **3 broadcast domain**.

## Packet Tracer - VLAN Configuration

### Addressing Table

| Device | Interface | IP Address   | Subnet Mask   | VLAN |
|--------|-----------|--------------|---------------|------|
| PC1    | NIC       | 172.17.10.21 | 255.255.255.0 | 10   |
| PC2    | NIC       | 172.17.20.22 | 255.255.255.0 | 20   |
| PC3    | NIC       | 172.17.30.23 | 255.255.255.0 | 30   |
| PC4    | NIC       | 172.17.10.24 | 255.255.255.0 | 10   |
| PC5    | NIC       | 172.17.20.25 | 255.255.255.0 | 20   |
| PC6    | NIC       | 172.17.30.26 | 255.255.255.0 | 30   |

### Objectives

**Part 1: Verify the Default VLAN Configuration**

**Part 2: Configure VLANs**

**Part 3: Assign VLANs to Ports**

### Background

VLANs are helpful in the administration of logical groups, allowing members of a group to be easily moved, changed, or added. This activity focuses on creating and naming VLANs, and assigning access ports to specific VLANs.

### Part 1: View the Default VLAN Configuration

#### Step 1: Display the current VLANs.

On S1, issue the command that displays all VLANs configured. By default, all interfaces are assigned to VLAN 1.

#### Step 2: Verify connectivity between PCs on the same network.

Notice that each PC can ping the other PC that shares the same subnet.

- PC1 can ping PC4
- PC2 can ping PC5
- PC3 can ping PC6

Pings to hosts on other networks fail.

What benefits can VLANs provide to the network?

VLAN dapat membagi jaringan menjadi beberapa kelompok secara logis. VLAN membantu meningkatkan keamanan, mengurangi broadcast traffic, dan memudahkan pengelolaan jaringan.

## Packet Tracer - VLAN Configuration

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### Part 2: Configure VLANs

#### Step 1: Create and name VLANs on S1.

- a. Create the following VLANs. Names are case-sensitive and must match the requirement exactly:
- VLAN 10: Faculty/Staff

```
S1#(config)# vlan 10
S1#(config-vlan)# name Faculty/Staff
```

- b. Create the remaining VLANs.
- VLAN 20: Students
  - VLAN 30: Guest(Default)
  - VLAN 99: Management&Native
  - VLAN 150: VOICE

#### Step 2: Verify the VLAN configuration.

Which command will only display the VLAN name, status, and associated ports on a switch?

Show vlan brief

#### Step 3: Create the VLANs on S2 and S3.

Use the same commands from Step 1 to create and name the same VLANs on S2 and S3.

#### Step 4: Verify the VLAN configuration.

### Part 3: Assign VLANs to Ports

#### Step 1: Assign VLANs to the active ports on S2.

- a. Configure the interfaces as access ports and assign the VLANs as follows:
- VLAN 10: FastEthernet 0/11

```
S2(config)# interface f0/11
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 10
```

- b. Assign the remaining ports to the appropriate VLAN.
  - VLAN 20: FastEthernet 0/18
  - VLAN 30: FastEthernet 0/6

## Step 2: Assign VLANs to the active ports on S3.

S3 uses the same VLAN access port assignments as S2. Configure the interfaces as access ports and assign the VLANs as follows:

- VLAN 10: FastEthernet 0/11
- VLAN 20: FastEthernet 0/18
- VLAN 30: FastEthernet 0/6

## Packet Tracer - VLAN Configuration

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## Step 3: Assign the VOICE VLAN to FastEthernet 0/11 on S3.

As shown in the topology, the S3 FastEthernet 0/11 interface connects to a Cisco IP Phone and PC4. The IP phone contains an integrated three-port 10/100 switch. One port on the phone is labeled Switch and connects to F0/4. Another port on the phone is labeled PC and connects to PC4. The IP phone also has an internal port that connects to the IP phone functions.

The S3 F0/11 interface must be configured to support user traffic to PC4 using VLAN 10 and voice traffic to the IP phone using VLAN 150. The interface must also enable QoS and trust the Class of Service (CoS) values assigned by the IP phone. IP voice traffic requires a minimum amount of throughput to support acceptable voice communication quality. This command helps the switchport to provide this minimum amount of throughput.

```
S3(config)# interface f0/11
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150
```

## Step 4: Verify loss of connectivity.

Previously, PCs that shared the same network could ping each other successfully.

Study the output of from the following command on **S2** and answer the following questions based on your knowledge of communication between VLANs. Pay close attention to the Gig0/1 port assignment.

```
S2# show vlan brief
```

| VLAN | Name              | Status | Ports                                                                                                                                                                                    |
|------|-------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default           | active | Fa0/1, Fa0/2, Fa0/3, Fa0/4<br>Fa0/5, Fa0/7, Fa0/8, Fa0/9<br>Fa0/10, Fa0/12, Fa0/13, Fa0/14<br>Fa0/15, Fa0/16, Fa0/17, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24, Gig0/1, Gig0/2 |
| 10   | Faculty/Staff     | active | Fa0/11                                                                                                                                                                                   |
| 20   | Students          | active | Fa0/18                                                                                                                                                                                   |
| 30   | Guest (Default)   | active | Fa0/6                                                                                                                                                                                    |
| 99   | Management&Native | active |                                                                                                                                                                                          |
| 150  | VOICE             | active | Try pinging between                                                                                                                                                                      |

PC1 and PC4.

Although the access ports are assigned to the appropriate VLANs, were the pings successful? Explain.

Tidak, ping tidak berhasil. Meskipun PC1 dan PC4 sama-sama berada di VLAN 10, port **Gig0/1** pada S2 masih berada di VLAN 1 dan belum dikonfigurasi sebagai trunk. Akibatnya, VLAN 10 tidak dapat diteruskan melalui koneksi antar-switch

What could be done to resolve this issue?

Konfigurasi port **Gig0/1** pada S2 dan port yang terhubung pada switch lainnya sebagai **trunk**, sehingga VLAN 10, VLAN 20, dan VLAN 30 dapat diteruskan antar-switch.

## Packet Tracer - Configure Trunks

### Addressing Table

| Device | Interface | IP Address   | Subnet Mask   | Switch Port | VLAN |
|--------|-----------|--------------|---------------|-------------|------|
| PC1    | NIC       | 172.17.10.21 | 255.255.255.0 | S2 F0/11    | 10   |
| PC2    | NIC       | 172.17.20.22 | 255.255.255.0 | S2 F0/18    | 20   |
| PC3    | NIC       | 172.17.30.23 | 255.255.255.0 | S2 F0/6     | 30   |
| PC4    | NIC       | 172.17.10.24 | 255.255.255.0 | S3 F0/11    | 10   |
| PC5    | NIC       | 172.17.20.25 | 255.255.255.0 | S3 F0/18    | 20   |
| PC6    | NIC       | 172.17.30.26 | 255.255.255.0 | S3 F0/6     | 30   |

### Objectives

#### Part 1: Verify VLANs

#### Part 2: Configure Trunks

### Background

Trunks are required to pass VLAN information between switches. A port on a switch is either an access port or a trunk port. Access ports carry traffic from a specific VLAN assigned to the port. A trunk port by default is a member of all VLANs. Therefore, it carries traffic for all VLANs. This activity focuses on creating trunk ports and assigning them to a native VLAN other than the default.

### Instructions

#### Part 1: Verify VLANs

##### Step 1: Display the current VLANs.

- a. On **S1**, issue the command that will display all VLANs configured. There should be ten VLANs in total. Notice that all 26 access ports on the switch are assigned to VLAN 1.
- b. On **S2** and **S3**, display and verify that all the VLANs are configured and assigned to the correct switch ports according to the **Addressing Table**.

### Step 2: Verify loss of connectivity between PCs on the same network.

Ping between hosts on the same VLAN on the different switches. Although **PC1** and **PC4** are on the same network, they cannot ping one another. This is because the ports connecting the switches are assigned to VLAN 1 by default. In order to provide connectivity between the PCs on the same network and VLAN, trunks must be configured.

## Part 2: Configure Trunks

### Step 1: Configure trunking on S1 and use VLAN 99 as the native VLAN.

### Packet Tracer - Configure Trunks

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- a. Configure G0/1 and G0/2 interfaces on S1 for trunking.

```
S1(config)# interface range g0/1 - 2 S1(config-if)# switchport mode trunk
```

- b. Configure VLAN 99 as the native VLAN for G0/1 and G0/2 interfaces on **S1**.

```
S1(config-if)# switchport trunk native vlan 99
```

The trunk port takes about a short time to become active due to Spanning Tree Protocol. Click **Fast Forward Time** to speed the process. After the ports become active, you will periodically receive the following syslog messages:

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/2 (99), with S3 GigabitEthernet0/2 (1).
```

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (99), with S2 GigabitEthernet0/1 (1).
```

You configured VLAN 99 as the native VLAN on S1. However, S2 and S3 are using VLAN 1 as the default native VLAN as indicated by the syslog message.

Although you have a native VLAN mismatch, pings between PCs on the same VLAN are now successful. Explain.

Karena setelah S1 dikonfigurasi sebagai trunk, trafik dari VLAN yang sama sudah dapat melewati trunk antara S1, S2, dan S3. Native VLAN yang berbeda hanya menimbulkan peringatan mismatch, tetapi tidak menghalangi trafik VLAN yang sudah diberi tag untuk melewati trunk.

## Step 2: Verify trunking is enabled on S2 and S3.

On **S2** and **S3**, issue the **show interface trunk** command to confirm that DTP has successfully negotiated trunking with S1 on S2 and S3. The output also displays information about the trunk interfaces on S2 and S3. You will learn more about DTP later in the course.

Which active VLANs are allowed to cross the trunk?

Semua VLAN aktif yang dikonfigurasi pada switch diperbolehkan melewati trunk secara default, termasuk **VLAN 1, 10, 20, 30, dan 99** serta VLAN aktif lainnya.

## Step 3: Correct the native VLAN mismatch on S2 and S3.

- a. Configure VLAN 99 as the native VLAN for the appropriate interfaces on S2 and S3.
- b. Issue **show interface trunk** command to verify the correct native VLAN configuration.

## Step 4: Verify configurations on S2 and S3.

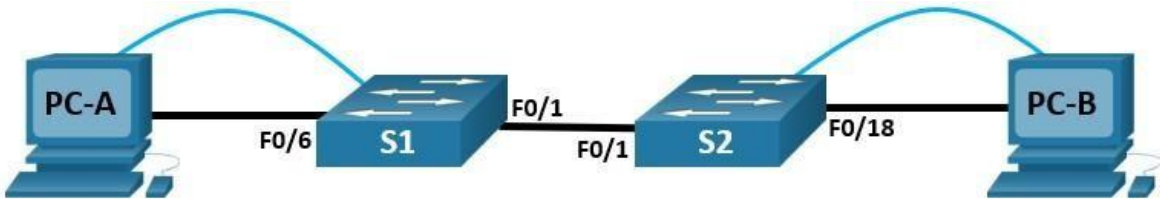
- a. Issue the **show interface *interface* switchport** command to verify that the native VLAN is now 99.
- b. Use the **show vlan** command to display information regarding configured VLANs.

Why is port G0/1 on S2 no longer assigned to VLAN 1?

Karena port **G0/1 pada S2 berubah menjadi trunk port**. Port trunk tidak lagi menjadi access port yang hanya berada di VLAN 1, tetapi dapat membawa trafik dari beberapa VLAN sekaligus. Native VLAN-nya juga telah diubah menjadi **VLAN 99**.

Packet Tracer - Configure VLANs and Trunking - Physical Mode

Topology



Addressing Table

|  | VLAN 1 | 192.168.1.11 | 255.255.255.0 | N/A          |
|--|--------|--------------|---------------|--------------|
|  | VLAN 1 | 192.168.1.12 | 255.255.255.0 | N/A          |
|  | NIC    | 192.168.10.3 | 255.255.255.0 | 192.168.10.1 |
|  | NIC    | 192.168.10.4 | 255.255.255.0 | 192.168.10.1 |

## Objectives

**Part 1: Build the Network and Configure Basic Device Settings**

**Part 2: Create VLANs and Assign Switch Ports**

**Part 3: Maintain VLAN Port Assignments and the VLAN Database Part**

**4: Configure an 802.1Q Trunk Between the Switches**

## Background / Scenario

Modern switches use virtual local-area networks (VLANs) to improve network performance by separating large Layer 2 broadcast domains into smaller ones. VLANs can also be used as a security measure by controlling which hosts can communicate. In general, VLANs make it easier to design a network to support the goals of an organization.

VLAN trunks are used to span VLANs across multiple devices. Trunks allow the traffic from multiple VLANs to travel over a single link, while keeping the VLAN identification and segmentation intact.

In this Packet Tracer Physical Mode (PTPM) activity, you will create VLANs on both switches in the topology, assign VLANs to switch access ports, and verify that VLANs are working as expected. You will then create a VLAN trunk between the two switches to allow hosts in the same VLAN to communicate through the trunk, regardless of which switch to which the host is attached.

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## Instructions

### Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings on the PC hosts and switches.

#### Step 1: Build the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary. a.

Click and drag both switch **S1** and **S2** to the **Rack**.

**Note:** This activity will open with 37% completion because the switch ports are all shutdown. When you install the switches in the rack, the ports will automatically be activated. After about a minute, the score will drop to 1%. Later in the activity, you will shut down unused ports.

- b. Click and drag both **PC-A** and **PC-B** to the **Table** and use the power button to turn them on.
- c. Provide network connectivity by connecting **Copper Straight-through** cables, as shown in the topology.
- d. Connect **Console Cable** from device **PC-A** to **S1** and from device **PC-B** to **S2**.

#### Step 2: Configure basic settings for each switch.

- a. From the **Desktop Tab** on each PC, use the **Terminal** to console into each switch and enable privileged EXEC mode.
- b. Enter configuration mode.
- c. Assign a device name to each switch.
- d. Assign **class** as the privileged EXEC encrypted password.
- e. Assign **cisco** as the console password and enable login.
- f. Assign **cisco** as the vty password and enable login.

- g. Encrypt the plaintext passwords.
- h. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- i. Configure the IP address listed in the Addressing Table for VLAN 1 on the switch.

**Note:** The VLAN 1 address is not graded because you will remove it later in the activity. However, you will need VLAN 1 to test connectivity later in this Part.

- j. Shut down all interfaces that will not be used.
- k. Set the clock on each switch.

**Note:** The clock setting cannot be graded in Packet Tracer.

- l. Save the running configuration to the startup configuration file.

### Step 3: Configure PC hosts.

From the **Desktop** tab on each **PC**, click IP Configuration and enter the addressing information as displayed in the Addressing Table.

### Step 4: Test connectivity.

Test network connectivity by attempting to ping between each of the cabled devices.

Can PC-A ping PC-B?

Ya, PC-A dapat melakukan ping ke PC-B karena keduanya masih berada pada VLAN 1 dan jaringan yang sama.

Can PC-A ping S1?

Ya, PC-A dapat melakukan ping ke S1 karena PC-A dan alamat IP VLAN 1 pada S1 berada dalam jaringan yang sama.

Can PC-B ping S2?

Ya, PC-B dapat melakukan ping ke S2 karena keduanya berada dalam jaringan yang sama.

Can S1 ping S2? Ya, S1 dapat melakukan ping ke S2 karena keduanya memiliki alamat IP

VLAN 1 dalam jaringan 192.168.1.0/24.

If you answered no to any of the

above questions, why were the pings unsuccessful?

Tidak ada yang gagal pada tahap ini karena semua perangkat masih menggunakan VLAN 1 dan berada dalam jaringan yang sesuai.

## Part 2: Create VLANs and Assign Switch Ports

In Part 2, you will create Management, Operations, Parking\_Lot, and Native VLANs on both switches. You will then assign the VLANs to the appropriate interface. The **show vlan** command is used to verify your configuration settings.

### Step 1: Create VLANs on the switches.

From the **Desktop Tab** on each **PC**, use Terminal to continue configuring both network switches.

- a. Create the VLANs on **S1**.  

```
S1(config)# vlan 10
S1(config-vlan)# name Operations
S1(config-vlan)# vlan 20
```

```

S1(config-vlan)# name Parking_Lot
S1(config-vlan)# vlan 99
S1(config-vlan)# name Management
S1(config-vlan)# vlan 1000
S1(config-vlan)# name Native
S1(config-vlan)# end

```

b. Create the same VLANs on S2.

c. Issue the **show vlan brief** command to view the list of VLANs on S1.

```
S1# show vlan brief
```

| VLAN | Name               | Status | Ports                                                                                                                                                                                                           |
|------|--------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/1, Fa0/2, Fa0/3, Fa0/4<br>Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gi0/1, Gi0/2 |
| 10   | Operations         | active |                                                                                                                                                                                                                 |
| 20   | Parking_Lot        | active |                                                                                                                                                                                                                 |
| 99   | Management         | active |                                                                                                                                                                                                                 |
| 1000 | Native             | active |                                                                                                                                                                                                                 |
| 1002 | fddi-default       | active |                                                                                                                                                                                                                 |
| 1003 | token-ring-default | active |                                                                                                                                                                                                                 |
| 1004 | fddinet-default    | active |                                                                                                                                                                                                                 |
| 1005 | trnet-default      | active |                                                                                                                                                                                                                 |

What is the default VLAN?

VLAN 1 adalah default VLAN pada switch.

What ports are assigned to the default VLAN?

What ports are assigned to the default VLAN?

Answer: Semua port switch yang belum dikonfigurasi berada di VLAN 1, yaitu Fa0/1 sampai Fa0/24 dan Gi0/1 sampai Gi0/2.

## Step 2: Assign VLANs to the correct switch interfaces.

a. Assign VLANs to the interfaces on S1.

1) Assign PC-A to the Operation VLAN.

```

S1(config)# interface f0/6
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10

```

2) From VLAN 1, remove the management IP address and configure it on VLAN 99.

```

S1(config)# interface vlan 1
S1(config-if)# no ip address

```

```
S1(config-if) # interface vlan 99  
S1(config-if) # ip address 192.168.1.11 255.255.255.0  
S1(config-if) # end
```

- b. Issue the **show vlan brief** command and verify that the VLANs are assigned to the correct interfaces. c. Issue the **show ip interface brief** command.

What is the status of VLAN 99? Explain.

VLAN 99 berstatus **up/up (aktif)** setelah digunakan sebagai interface VLAN untuk alamat IP management. VLAN 99 dapat aktif karena terdapat port yang aktif dan ditetapkan ke VLAN 99.

- d. Assign **PC-B** to the Operations VLAN on **S2**.  
e. From VLAN 1, remove the management IP address and configure it on VLAN 99 according to the Addressing Table .  
f. Use the **show vlan brief** command to verify that the VLANs are assigned to the correct interfaces.

Is S1 able to ping S2? Explain.

Tidak, pada tahap ini S1 belum dapat melakukan ping ke S2 melalui VLAN 99 karena link antara kedua switch masih menggunakan VLAN 1 dan belum dikonfigurasi sebagai trunk.

Is PC-A able to ping PC-B? Explain.

Tidak, PC-A dan PC-B sudah berada pada VLAN 10, tetapi koneksi antar-switch belum menggunakan trunk. Akibatnya, trafik VLAN 10 belum dapat melewati link antara S1 dan S2.

### Part 3: Maintain VLAN Port Assignments and the VLAN Database

In Part 3, you will change port VLAN assignments and remove VLANs from the VLAN database.

#### Step 1: Assign a VLAN to multiple interfaces.

From the **Desktop Tab** on each **PC**, use **Terminal** to continue configuring both network switches.

- a. On S1, assign interfaces F0/11 – 24 to VLAN99.  
S1(config) # **interface range f0/11-24**  
S1(config-if-range) # **switchport mode access**  
S1(config-if-range) # **switchport access vlan 99**  
S1(config-if-range) # **end**
- b. Issue the **show vlan brief** command to verify VLAN assignments.  
c. Reassign F0/11 and F0/21 to VLAN 10.  
d. Verify that VLAN assignments are correct.

#### Step 2: Remove a VLAN assignment from an interface.

- a. Use the **no switchport access vlan** command to remove the VLAN 99 assignment to F0/24.  
S1(config) # **interface f0/24**  
S1(config-if) # **no switchport access vlan**  
S1(config-if) # **end**

- b. Verify that the VLAN change was made.

Which VLAN is F0/24 now associated with?

F0/24 kembali terkait dengan **VLAN 1**, karena perintah `no switchport access vlan` menghapus assignment VLAN 99 dan mengembalikannya ke VLAN default

### Step 3: Remove a VLAN ID from the VLAN database.

- a. Add VLAN 30 to interface F0/24 without issuing the global VLAN command.

```
S1(config)# interface f0/24
```

```
S1(config-if)# switchport access vlan 30
```

```
% Access VLAN does not exist. Creating vlan 30
```

**Note:** Current switch technology no longer requires that the **vlan** command be issued to add a VLAN to the database. By assigning an unknown VLAN to a port, the VLAN will be created and added to the VLAN database.

- b. Verify that the new VLAN is displayed in the VLAN table.

What is the default name of VLAN 30?

Nama default VLAN 30 adalah **VLAN0030**.

- c. Use the **no vlan 30** command to remove VLAN 30 from the VLAN database.

```
S1(config)# no vlan 30
```

```
S1(config)# end
```

- d. Issue the **show vlan brief** command. F0/24 was assigned to VLAN 30.

After deleting VLAN 30 from the VLAN database, why is F0/24 no longer displayed in the output of the **show vlan brief** command? What VLAN is port F0/24 now assigned to? What happens to the traffic destined to the host that is attached to F0/24?

F0/24 tidak lagi ditampilkan karena VLAN 30 sudah dihapus dari VLAN database. Port F0/24 menjadi tidak aktif secara fungsional untuk VLAN tersebut sampai assignment VLAN diperbaiki. Traffic yang ditujukan ke host pada F0/24 tidak dapat diteruskan dengan normal karena VLAN 30 sudah tidak ada.

- e. On interface F0/24, issue the **no switchport access vlan** command.

- f. Issue the **show vlan brief** command to determine the VLAN assignment for F0/24.

To which VLAN is F0/24 assigned?

: F0/24 kembali ditetapkan ke **VLAN 1** setelah perintah `no switchport access vlan` digunakan.

**Note:** Before removing a VLAN from the database, it is recommended that you reassign all the ports assigned to that VLAN.

Why should you reassign a port to another VLAN before removing the VLAN from the VLAN database?

Agar port tidak kehilangan VLAN yang valid dan komunikasi perangkat yang terhubung ke port tersebut tetap dapat berjalan. Sebaiknya port dipindahkan terlebih dahulu ke VLAN lain sebelum VLAN lama dihapus.

## Part 4: Configure an 802.1Q Trunk Between the Switches

In Part 4, you will configure interface F0/1 to use the Dynamic Trunking Protocol (DTP) to allow it to negotiate the trunk mode. After this has been accomplished and verified, you will disable DTP on interface F0/1 and manually configure it as a trunk.

### Step 1: Use DTP to initiate trunking on F0/1.

The default DTP mode of a 2960 switch port is dynamic auto. This allows the interface to convert the link to a trunk if the neighboring interface is set to trunk or dynamic desirable mode.

- a. On **S1**, set F0/1 to negotiate trunk mode.

```
S1(config)# interface f0/1
S1(config-if)# switchport mode dynamic desirable
Sep 19 02:51:47.257: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Sep 19 02:51:47.291: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
```

You should also receive link status messages on S2.

```
S2#
Sep 19 02:42:19.424: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
Sep 19 02:42:21.454: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
Sep 19 02:42:22.419: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

- b. On **S1** and **S2**, issue the **show vlan brief** command. Interface F0/1 is no longer assigned to VLAN 1. Trunked interfaces are not listed in the VLAN table.

- c. Issue the **show interfaces trunk** command to view trunked interfaces. Notice that the mode on **S1** is set to desirable, and the mode on **S2** is set to auto.

```
S1# show interfaces trunk
```

```
S2# show interfaces trunk
```

**Note:** By default, all VLANs are allowed on a trunk. The **switchport trunk** command allows you to control what VLANs have access to the trunk. For this activity, keep the default settings. This allows all VLANs to traverse F0/1.

- d. Verify that VLAN traffic is traveling over trunk interface F0/1.

Can S1 ping S2?

Ya, S1 dapat melakukan ping ke S2 karena trunk F0/1 sudah terbentuk dan dapat membawa trafik VLAN 99.

Can PC-A ping PC-B?

Ya, PC-A dapat melakukan ping ke PC-B karena keduanya berada pada VLAN 10 dan VLAN 10 dapat melewati trunk F0/1.

Can PC-A ping S1?

Ya, PC-A dapat melakukan ping ke S1 karena PC-A berada pada VLAN 10 dan konektivitas antar-switch sudah tersedia melalui trunk.

Can PC-B ping S2?

Ya, PC-B dapat melakukan ping ke S2 karena PC-B terhubung pada VLAN 10 dan trunk memungkinkan trafik VLAN tersebut melewati kedua switch.

If you answered no to any of the above questions, explain below.

## Step 2: Manually configure trunk interface F0/1.

The **switchport mode trunk** command is used to manually configure a port as a trunk. This command should be issued on both ends of the link.

- a. On interface F0/1, change the switchport mode to force trunking. Make sure to do this on both switches.

```
S1(config)# interface f0/1
```

```
S1(config-if)# switchport mode trunk
```

- b. Issue the **show interfaces trunk** command to view the trunk mode. Notice that the mode changed from **desirable** to **on**.

```
S1# show interfaces trunk
```

- c. Modify the trunk configuration on both switches by changing the native VLAN from VLAN 1 to VLAN 1000.

```
S1(config)# interface f0/1
```

```
S1(config-if)# switchport trunk native vlan 1000
```

- d. Issue the **show interfaces trunk** command to view the trunk. Notice the Native VLAN information is updated.

```
S2# show interfaces trunk
```

Why might you want to manually configure an interface to trunk mode instead of using DTP?

Untuk mengontrol konfigurasi trunk secara manual dan meningkatkan keamanan serta menghindari perubahan mode trunk yang tidak diinginkan melalui DTP.

Why might you want to change the native VLAN on a trunk?

Untuk meningkatkan keamanan jaringan dan mengurangi risiko serangan yang memanfaatkan native VLAN default. Pada lab ini native VLAN diubah dari VLAN 1 menjadi VLAN 1000

## Reflection Questions

1. What is needed to allow hosts on VLAN 10 to communicate to hosts on VLAN 99?

Diperlukan **routing antar-VLAN (Inter-VLAN Routing)** menggunakan router atau Layer 3 switch, karena VLAN 10 dan VLAN 99 merupakan broadcast domain yang berbeda

2. What are some primary benefits that an organization can receive through effective use of VLANs?

VLAN dapat meningkatkan **keamanan, mengurangi broadcast traffic, meningkatkan performa jaringan, serta mempermudah pengelolaan dan pembagian jaringan secara logis**. Materi juga menjelaskan bahwa VLAN dapat digunakan untuk mengontrol host mana yang dapat berkomunikasi.

## Packet Tracer - Configure DTP

### Addressing Table

| Device | Interface | IP Address   | Subnet Mask   |
|--------|-----------|--------------|---------------|
| PC1    | NIC       | 192.168.10.1 | 255.255.255.0 |
| PC2    | NIC       | 192.168.20.1 | 255.255.255.0 |
| PC3    | NIC       | 192.168.30.1 | 255.255.255.0 |
| PC4    | NIC       | 192.168.30.2 | 255.255.255.0 |
| PC5    | NIC       | 192.168.20.2 | 255.255.255.0 |
| PC6    | NIC       | 192.168.10.2 | 255.255.255.0 |
| S1     | VLAN 99   | 192.168.99.1 | 255.255.255.0 |
| S2     | VLAN 99   | 192.168.99.2 | 255.255.255.0 |
| S3     | VLAN 99   | 192.168.99.3 | 255.255.255.0 |

### Objectives

- Configure static trunking
- Configure and Verify DTP

### Background / Scenario

As the number of switches in a network increases, the administration necessary to manage the VLANs and trunks can be challenging. To ease some of the VLAN and trunking configurations, trunk negotiation between network devices is managed by the Dynamic Trunking Protocol (DTP), and is automatically enabled on Catalyst 2960 and Catalyst 3650 switches.

In this activity, you will configure trunk links between the switches. You will assign ports to VLANs and verify end-to-end connectivity between hosts in the same VLAN. You will configure trunk links between the switches, and you will configure VLAN 999 as the native VLAN.

### Instructions

#### Part 1: Verify VLAN configuration.

Verify the configured VLANs on the switches.

- On S1, go to privileged EXEC mode and enter the **show vlan brief** command to verify the VLANs that are present.

```
S1# show vlan brief
```

| VLAN | Name               | Status | Ports                                                                                                                                                                                                             |
|------|--------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/1, Fa0/2, Fa0/3, Fa0/4<br>Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2 |
| 99   | Management         | active |                                                                                                                                                                                                                   |
| 999  | Native             | active |                                                                                                                                                                                                                   |
| 1002 | fddi-default       |        | active                                                                                                                                                                                                            |
| 1003 | token-ring-default |        | active                                                                                                                                                                                                            |
| 1004 | fddinet-default    |        | active                                                                                                                                                                                                            |
| 1005 | trnet-default      |        | active                                                                                                                                                                                                            |

- b. Repeat Step 1a on S2 and S3.

What VLANs are configured on the switches?

VLAN yang dikonfigurasi adalah **VLAN 1 (default)**, **VLAN 99 (Management)**, dan **VLAN 999 (Native)**.

## Part 2: Create additional VLANs on S2 and S3.

- a. On S2, create VLAN 10 and name it Red.

```
S2(config)# vlan 10
```

```
S2(config-vlan)# name Red
```

- b. Create VLANs 20 and 30 according to the table below.

| VLAN Number | VLAN Name |
|-------------|-----------|
| 10          | Red       |
| 20          | Blue      |
| 30          | Yellow    |

- c. Verify the addition of the new VLANs. Enter **show vlan brief** at the privileged EXEC mode.

In addition to the default VLANs, which VLANs are configured on S2

Selain VLAN default, S2 memiliki **VLAN 10 (Red)**, **VLAN 20 (Blue)**, dan **VLAN 30 (Yellow)**.

- d. Repeat the previous steps to create the additional VLANs on S3.

## Part 3: Assign VLANs to Ports

Use the **switchport mode access** command to set access mode for the access links. Use the **switchport access vlan** *vlan-id* command to assign a VLAN to an access port.

| Ports                                      | Assignments         | Network             |
|--------------------------------------------|---------------------|---------------------|
| S2<br>F0/1<br>– 8<br>S3<br>F0/1<br>– 8     | VLAN 10<br>(Red)    | 192.168.10.0<br>/24 |
| S2<br>F0/9<br>– 16<br>S3<br>F0/9<br>– 16   | VLAN 20<br>(Blue)   | 192.168.20.0<br>/24 |
| S2<br>F0/17<br>– 24<br>S3<br>F0/17<br>– 24 | VLAN 30<br>(Yellow) | 192.168.30.0<br>/24 |

a. Assign VLANs to ports on S2 using assignments from the table above.

```
S2(config-if)# interface range f0/1 - 8
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 10
S2(config-if-range)# interface range f0/9 -16
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 20
S2(config-if-range)# interface range f0/17 - 24
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 30
```

b. Assign VLANs to ports on S3 using the assignments from the table above.

Now that you have the ports assigned to VLANs, try to ping from **PC1** to **PC6**.

Was the ping successful? Explain.

Tidak, ping dari PC1 ke PC6 tidak berhasil. PC1 dan PC6 memang berada pada jaringan yang samasama sesuai dengan VLAN 10, tetapi VLAN belum dapat melewati koneksi antar-switch karena link antar-switch belum dikonfigurasi sebagai trunk.

## Part 4: Configure Trunks on S1, S2, and S3.

Dynamic trunking protocol (DTP) manages the trunk links between Cisco switches. Currently, all the switchports are in the default trunking mode, which is dynamic auto. In this step, you will change the trunking mode to dynamic desirable for the link between switches S1 and S2. The link between switches S1 and S3 will be set as a static trunk. Use VLAN 999 as the native VLAN in this topology.

- a. On switch S1, configure the trunk link to dynamic desirable on the GigabitEthernet 0/1 interface. The configuration of S1 is shown below.

```
S1(config)# interface g0/1
S1(config-if)# switchport mode dynamic desirable
```

What will be the result of trunk negotiation between S1 and S2?

Trunk akan berhasil dinegosiasikan karena S1 menggunakan mode **dynamic desirable**, sedangkan S2 masih menggunakan mode **dynamic auto**. Kombinasi tersebut akan membuat link antara S1 dan S2 menjadi trunk.

---

- b. On switch S2, verify that the trunk has been negotiated by entering the **show interfaces trunk** command. Interface GigabitEthernet 0/1 should appear in the output.

What is the mode and status for this port?

Mode port adalah **auto** dan statusnya **trunking**. S2 menggunakan dynamic auto dan berhasil menjadi trunk karena S1 menggunakan dynamic desirable.

- c. For the trunk link between S1 and S3, configure interface GigabitEthernet 0/2 as a static trunk link on S1. In addition, disable DTP negotiation on interface G0/2 on S1.

```
S1(config)# interface g0/2
S1(config-if)# switchport mode trunk
S1(config-if)# switchport nonegotiate
```

- d. Use the **show dtp** command to verify the status of DTP.

```
S1# show dtp
Global DTP information
  Sending DTP Hello packets every 30 seconds
  Dynamic Trunk timeout is 300 seconds
  1 interfaces using DTP
```

- e. Verify trunking is enabled on all the switches using the **show interfaces trunk** command.

```
S1# show interfaces trunk
```

| Port   | Mode      | Encapsulation | Status   | Native vlan |
|--------|-----------|---------------|----------|-------------|
| Gig0/1 | desirable | n-802.1q      | trunking | 1           |
| 802.1q | trunking  | 1             |          |             |

| Port   | Vlans allowed on trunk |
|--------|------------------------|
| Gig0/1 | 1-1005                 |
| Gig0/2 | 1-1005                 |

| Port   | Vlans allowed and active in management domain |
|--------|-----------------------------------------------|
| Gig0/1 | 1,99,999                                      |
| Gig0/2 | 1,99,999                                      |

| Port   | Vlans in spanning tree forwarding state and not pruned |
|--------|--------------------------------------------------------|
| Gig0/1 | 1,99,999                                               |
| Gig0/2 | 1,99,999                                               |

What is the native VLAN for these trunks currently?

Native VLAN untuk trunk tersebut saat ini adalah **VLAN 1**.

f. Configure VLAN 999 as the native VLAN for the trunk links on S1.

```
S1(config)# interface range g0/1 - 2
```

```
S1(config-if-range)# switchport trunk native vlan 999
```

---

What messages did you receive on S1? How would you correct it?

Muncul pesan **Native VLAN mismatch** karena native VLAN pada S1 sudah diubah menjadi VLAN 999, sedangkan S2 dan S3 masih menggunakan VLAN 1. Cara memperbaikinya adalah mengatur native VLAN menjadi **VLAN 999 pada S2 dan S3** juga.

g. On S2 and S3, configure VLAN 999 as the native VLAN.

h. Verify trunking is successfully configured on all the switches. You should be able ping one switch from another switch in the topology using the IP addresses configured on the SVI.

i. Attempt to ping from PC1 to PC6.

Why was the ping unsuccessful? (Hint: Look at the '**show vlan brief**' output from all three switches.

Compare the outputs from the '**show interface trunk**' on all switches.) j.

Correct the configuration as necessary.

## Part 5: Reconfigure trunk on S3.

a. Issue the '**show interface trunk**' command on S3.

What is the mode and encapsulation on G0/2?

Mode G0/2 pada S3 adalah **dynamic auto** dan encapsulation-nya **802.1Q**.

b. Configure G0/2 to match G0/2 on S1.

What is the mode and encapsulation on G0/2 after the change?

Setelah dikonfigurasi agar sama dengan S1, mode G0/2 menjadi **on** dan encapsulation-nya **802.1Q**.

Konfigurasinya:

```
S3(config)# interface g0/2
```

```
S3(config-if)# switchport mode trunk
```

```
S3(config-if)# switchport nonegotiate
```

- c. Issue the command '**show interface G0/2 switchport**' on switch **S3**.

What is the '**Negotiation of Trunking**' state displayed?

Status **Negotiation of Trunking** adalah **Off**, karena perintah switchport nonegotiate digunakan untuk menonaktifkan DTP.

## **Part 6: Verify end to end connectivity.**

- a. From PC1 ping PC6.
- b. From PC2 ping PC5.
- c. From PC3 ping PC4.

**Packet Tracer - Implement VLANs and Trunking  
Addressing Table**

| Device | Interface | IP Address     | Subnet Mask   | Switchport | VLAN                          |
|--------|-----------|----------------|---------------|------------|-------------------------------|
| PC1    | NIC       | 192.168.10.10  | 255.255.255.0 | SWB F0/1   | VLAN 10                       |
| PC2    | NIC       | 192.168.20.20  | 255.255.255.0 | SWB F0/2   | VLAN 20                       |
| PC3    | NIC       | 192.168.30.30  | 255.255.255.0 | SWB F0/3   | VLAN 30                       |
| PC4    | NIC       | 192.168.10.11  | 255.255.255.0 | SWC F0/1   | VLAN 10                       |
| PC5    | NIC       | 192.168.20.21  | 255.255.255.0 | SWC F0/2   | VLAN 20                       |
| PC6    | NIC       | 192.168.30.31  | 255.255.255.0 | SWC F0/3   | VLAN 30                       |
| PC7    | NIC       | 192.168.10.12  | 255.255.255.0 | SWC F0/4   | VLAN 10<br>VLAN 40<br>(Voice) |
| SWA    | SVI       | 192.168.99.252 | 255.255.255.0 | N/A        | VLAN 99                       |
| SWB    | SVI       | 192.168.99.253 | 255.255.255.0 | N/A        | VLAN 99                       |
| SWC    | SVI       | 192.168.99.254 | 255.255.255.0 | N/A        | VLAN 99                       |

**Objectives**

**Part 1: Configure VLANs**

**Part 2: Assign Ports to VLANs**

**Part 3: Configure Static Trunking**

**Part 4: Configure Dynamic Trunking**

**Background**

You are working in a company that is getting ready to deploy a set of new 2960 switches in a branch office. You are working in the lab to test out the VLAN and trunking configurations that are planned. Configure and test the VLANs and trunks. **Instructions**

### Part 1: Configure VLANs

Configure VLANs on all three switches. Refer to the VLAN Table. Note that the VLAN names must match the values in the table exactly.

#### VLAN Table

| VLAN Number | VLAN Name |
|-------------|-----------|
| 10          | Admin     |

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### Packet Tracer - Implement VLANs and Trunking

| VLAN Number | VLAN Name  |
|-------------|------------|
| 20          | Accounts   |
| 30          | HR         |
| 40          | Voice      |
| 99          | Management |
| 100         | Native     |

### Part 2: Assign Ports to VLANs

#### Step 1: Assign access ports to VLANs

On SWB and SWC, assign ports to the VLANs. Refer to the Addressing Table. **Step 2: Configure the Voice VLAN port**

Configure the appropriate port on switch SWC for voice VLAN functionality.

#### Step 3: Configure the virtual management interfaces

- Create the virtual management interfaces, on all three switches.
- Address the virtual management interfaces according to the Addressing Table.
- The switches should not be able to ping each other.

### Part 3: Configure Static Trunking

- Configure the link between SWA and SWB as a static trunk. Disable dynamic trunking on this port.
- Disable DTP on the switch port on both ends of the trunk link.
- Configure the trunk with the native VLAN and eliminate native VLAN conflicts if any.

### Part 4: Configure Dynamic Trunking

- Assume that the trunk port on SWC is set to the default DTP mode for 2960 switches. Configure G0/2 on SWA so that it successfully negotiates trunking with SWC.
- Configure the trunk with the native VLAN and eliminate native VLAN conflicts if any.

*End of document*

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